

Try Your Hand at GNOME Shell Extensions

One of the most popular features of GNOME is that it enables users to build extensions for their shell to increase its basic functionality. You can think of these extensions as add-ons to the GNOME shell. Here's more on developing GNOME shell extensions.

One of the many choices available to a Linux user is that of the desktop environment. Among the many available, GNOME stands out as one of the most prominent. The GNOME shell has a very user friendly interface, which makes it popular amongst users.

First things first. GNOME extensions are written in Javascript. This is actually Mozilla's special version of Javascript, which is based on Spidermonkey and is widely known as GJS. A working knowledge of regular Javascript is enough to begin extension development.

Let us have a look at what the components of a GNOME extension are.

- **Metadata.json:** This file contains data about the extension—details like the extension name, description, identifier and compatibility information (which versions of the GNOME shell are compatible with the extension).
- **Extension.js:** This is the GNOME-Javascript(GJS) file, which will store the logic that controls the extension. This file must contain the following three functions.
 - **Init (metadata)** – This is called to initialise the extension. The metadata argument is the data stored in the *metadata.json* file, along with some additional details like the absolute path to your extension.
 - **enable** – This is called when the extension is enabled. This function stores the actions to be performed when the extension is enabled, and this generally is the function in which the logic for extensions is placed.
 - **disable** – This is called when the extension is disabled. It generally contains code, related to clean up activities, which stops whatever actions the extensions perform and restores the system to a state such as when the extension was never there.

There is also an optional *stylesheet.css* file, which is used to style various attributes of the extension, like buttons or displays.

Creating an extension

When you write your extension, the files related to the extension should be placed in the `~/local/share/gnome-shell/extensions` directory. In the same directory, you will be able to see various other extensions that exist on your system, and it may be a good idea to look at some of them to get a better understanding of how they work. You will also see that the already existing extensions will have the above mentioned files (*extension.js*, *stylesheet.css*, *metadata.json*).

Now, let us start building our extension. Although it is not mandatory (you can write the extension from scratch),



GNOME 3 has a bash command that helps create new extensions. Let's use the following command to create a draft of our extension:

```
gnome-shell-extension-tool --create-extension
```

When you run the above command, you will be prompted to enter the following metadata.

- **Name** – Name of the extension.
- **Description** – Description of the extension.
- **UUID** – This is a unique identifier for the extension. It should be in an email address format, like `foo@bar.com`. In general, as long as the UUID can be used to create a subdirectory, it will be regarded as a valid UUID.

Now, you will be told that your extension has been created and the path to it will also be displayed. You are free to make the required changes to your extension. Next, activate the extension by using the GNOME tweak-tool (if you have not already installed it, you can download and install it from your distro's software manager). Under the tweak tool, go to shell extensions and turn on your extension there. If you cannot see it, restart your GNOME shell (Alt + F2 and type 'r', and then press 'Enter') to force load all the extensions.

You will see a new icon in the GNOME tray (top right corner of your screen) in the shape of gears. If you click on it, your extension will display 'Hello world'.

This is the default extension, created by the GNOME shell extension tool. We must modify the files located in `~/local/share/gnome-shell/extensions`, in the folder with the same name as the UUID you entered earlier.

Understanding the code

Let's look at the code that the GNOME shell extension-tool created, by first examining the *extension.js* file: